Objects as Conditions: Object-Centric State and Relational Composition for Pretrained Vision-Language-Action Policies

Ali Younes[1]

co-authors, affiliations, funding

***Abstract*—Pretrained vision-language-action (VLA) policies read the scene from pixels. We ask whether giving them an explicit object-centric description of the scene helps them manipulate. The description is an unordered set of entities with pose, size and type. We also ask whether *how the entities are composed* matters. On a dual-arm, multi-object restocking task, zero-initialised, query-dependent per-entity attention roughly quadruples the task success of GR00T N1.7 over an RGB policy given the same training budget. It goes from 9 to 35 of 60 scenes with seen objects, and from 0 to 11 of 60 with held-out object sizes. Two controls show that the gain is object information and not added capacity. With shuffled entities the policy falls back to the control's level, and a pooled scene vector alone does not help. The effect depends on the backbone: ACT gains only on held-out sizes, and pi0.5 and SmolVLA gain nothing once capacity is matched. We then test a simple relational composition rule, in which pairwise relative geometry is passed between entities before injection. We test it on the restocking task and five RoboTwin 2.0 multi-object tasks. Finally, we test generalisation across object count by training on three objects and evaluating on two and four.**

Rewrite once RQ2/RQ3 are in. Keep under 250 words.

***Index Terms*—Imitation Learning, Deep Learning in Grasping and Manipulation, Bimanual Manipulation, Representation Learning.**

Open items (remove before submission)

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- Paragraph plan: (1) problem: a VLA fine-tuned from RGB must recover object identity and relative geometry from pixels, from few demonstrations; (2) gap: ControlVLA injects objects as a flat set, and nobody has checked, against a budget-matched control and across backbones, whether that helps, which injection carries the gain, or whether relational composition helps further; (3) findings so far; (4) the proposal: a relational encoder and a semantic channel; (5) the evaluation protocol. . . . 1
- Teaser caption. . . . 2
- Demos per task; oracle success rate (ceiling) per task. 2
- Open: why GR00T but not $\pi_{0.5}$/SmolVLA? GR00T's branch is ˜150M parameters against ˜1M for the matched LoRA branch, so the ratio is the same but the absolute size is not. Decide whether to test a larger branch with a capacity-matched control, or to report the backbone dependence as a finding. . . . 3
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- LPWM: authors and title . . . 5

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[1]Affiliation.

I. Introduction

MULTI-OBJECT manipulation requires knowing which object to act on, where it is, and where it is relative to the others.

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Claim. Object state helps a pretrained VLA when each action query retrieves it per entity, not when it is summarised. The next step is to give that retrieval the relations between objects.

Contributions:

1) [DONE] A controlled study of object-centric conditioning across four policy families (ACT, GR00T N1.7, $\pi_{0.5}$, SmolVLA). Every arm is paired against a training-budget-matched RGB control, with three training seeds per cell (Sec. V).
2) [DONE] Evidence that the GR00T gain is object *information* and that *query-dependent* composition carries it. It is also robust to 2–5 mm pose noise and to visual appearance shift (Sec. V).
3) [TO RUN] A relational composition module that starts as the KV policy, together with information and capacity controls for it, and a semantic (colour) channel (Sec. VI).

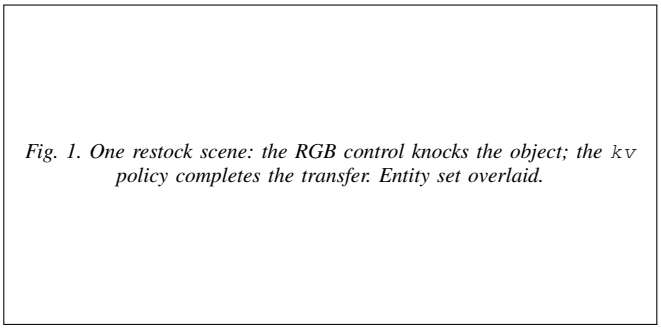


Fig. 1. One restock scene: the RGB control knocks the object; the `kv` *policy completes the transfer. Entity set overlaid.*

Fig. 1.
Teaser caption.

4) [TO RUN] Generalisation across object count (train on 3, test on 2 and 4) for every arm (Sec. VII). The existing arms are [DONE].

II. Related Work

VLA policies and adaptation. ACT [1], GR00T N1 [2], $\pi_{0.5}$ [3], SmolVLA [4]. These models adapt from pixels, and adaptation methods are rarely compared against a budget-matched control.

Object-centric conditioning. ControlVLA [5] uses zero-initialised KV control, and its "without pretraining" ablation is one we replicate. Related work includes VIOLA [7], slot representations [8], and LPWM [6], which conditions by AdaLN or by tokens. We hold the representation fixed and vary the injection and the composition.

Relational inductive biases. Relevant work includes relation networks [9], graph networks [10] and set functions [11]. We use the simplest relational layer that leaves the policy's behaviour at step 0 unchanged.

Benchmarks. RoboTwin 2.0 [12].

III. Method

A. The entity set

A scene is an unordered set of up to $N{=}16$ entities: the movable objects, both grippers and the support surfaces, with a mask marking which rows are real. Entity i is $z_i = (p_i, r_i, s_i, a_i, \tau_i)$. Here p_i is the position in the workcell frame, r_i the 6D rotation, s_i the size, a_i the gripper aperture and τ_i a one-hot type over {object, left gripper, right gripper, support}, giving 17 columns. The set carries no task role, target id or object id. Position and size are standardised with statistics from the training split only. Entities come from the simulator's ground truth. We treat this as an upper bound on what perception can deliver, and measure robustness to noise and dropout in Sec. V.

B. Two-stage recipe and injection

Stage 1 trains the stock policy on RGB and proprioception. *Stage 2* loads that checkpoint and adds the conditioning branch, initialised so that the stage-2 policy is numerically the stage-1 policy at step 0. This is verified bit-for-bit on ACT and by an init check on every VLA.

- `kv`: in every hooked host attention layer, the output becomes

$$W_o\left[\mathrm{sm}\left(\frac{QK^\top}{\sqrt{d}}\right)V + \mathrm{sm}\left(\frac{QK_z^\top}{\sqrt{d}}\right)V_z\right], \quad (1)$$

  where $K_z{=}W_k h$ and $V_z{=}W_v h$, with W_k, W_v initialised to zero and $h_i = E(z_i)$ (ControlVLA [5], with masking and no positional code).
- `kv_adaln`: `kv` plus an attention-pooled scene vector that modulates every block (AdaLN [6]).

C. Relational composition (proposed)

The relational arm, `kv_rel`, replaces $h_i = E(z_i)$ with a relation-aware embedding. For each pair it expresses j in i's frame:

$$u_{ij} = \left[R_i^\top(p_j - p_i),\ 6\mathrm{D}(R_i^\top R_j),\ \|p_j - p_i\|,\ s_j - s_i\right], \quad (2)$$

$$h_i = E(z_i) + W_0 \frac{1}{|\mathcal{N}_i|}\sum_{j\in\mathcal{N}_i} \phi\big(E(z_i), E(z_j), \psi(u_{ij})\big), \quad (3)$$

where $\mathcal{N}_i$ is the set of real entities $j \neq i$. The mean makes the aggregate independent of how many objects there are, which is what count generalisation tests. The zero-initialised W_k, W_v already make the arm identical to stage 1 at step 0. W_0 is therefore *not* also zero-initialised. If it were, ϕ would receive no gradient until W_v moved, and the init check asserts that gradient reaches ϕ. The default is one layer, with two layers as an ablation. Gripper-relative terms ($i \in$ grippers) address the residual ~9 mm horizontal grasp error, which no absolute-pose arm reduces (Sec. V).

D. Semantic channel (proposed)

Two RoboTwin tasks define their goal by colour (Table I), which no geometric entity can express. The `+c` variants append each entity's base colour (RGB, 3 columns), declared per task binding. This replaces the current 32-column "semantic" entity, whose semantic half hashes the object's size and so only re-encodes geometry.

IV. Experimental Setup

Restock (ours). A dual-arm Panda in SAPIEN moves three objects from a lower to an upper shelf. The objects come from seven mesh sizes:

- seen: sizes 1–4;
- held-out: sizes 0 and 6, collected but excluded from training;
- novel: size 5, which the oracle never collected.

Training uses 75 oracle runs recorded as full continuous runs. Evaluation counts are also run with 2 and 4 objects, on a 2-object layout nested inside the 3-object one.

RoboTwin 2.0. Five multi-object tasks (Table I), each with its built-in oracle and success checker. [TO RUN]
Demos per task; oracle success rate (ceiling) per task.

Backbones.

- ACT: 40k steps per stage, batch 8, chunk 50, executing 25.

TABLE I
ROBOTWIN TASKS AND THE RELATION EACH GOAL REQUIRES.

Task	Movable entities	Goal relation	Colour
stack_blocks_three	3 blocks	stacked R<G<B	✓
blocks_ranking_rgb	3 blocks	ordered R,G,B in x	✓
put_bottles_dustbin	3 bottles + bin	containment, sequence	–
place_cans_plasticbox	2 cans + box	bimanual containment	–
place_bread_basket	2 breads + basket	containment	–

- GR00T N1.7: frozen backbone, head trained for 8k steps per stage at batch 16.
- $\pi_{0.5}$ and SmolVLA: action-expert LoRA (rank 32, VLM frozen), with the object branch held to GR00T's share of trainable parameters (+8–19%).

Arms and controls.

- `rgb_cont` continues stage 1 for as many steps as stage 2 adds. It is the fair baseline and every arm is paired against it.
- `scratch_kv` is `kv` trained without stage 1.

Protocol and statistics.

- Every cell is 3 training seeds × 20 pinned scenes (seeds 800–819).
- Episodes run 600 steps for 3 objects and 200 steps per object otherwise.
- Metrics: task success, at least one transfer, mean transfers, and stage-wise survival, each read against the oracle ceiling.
- Arms are compared by exact McNemar on matched (training seed, scene) episodes. Many contrasts are tested, so $p < 0.05$ counts as a lead, and a claim requires the same sign on every seed.
- Validation loss is never used to rank arms or pick checkpoints: it contradicted closed-loop success in nine documented cases.

V. RQ1: IS OBJECT-CENTRIC STATE USEFUL?

Claim. Yes for GR00T, on seen and held-out objects. For ACT, only on held-out sizes and only through AdaLN. For $\pi_{0.5}$ and SmolVLA, no, at matched capacity.

Have. Restock, every backbone, 3 seeds (Tables II, III); `eval/{CF,GF,PXF,SXF}-*.json`.

Need. RoboTwin: `rgb`, `rgb_cont`, `kv`, `kv_adaln` × ACT, GR00T × 5 tasks × 3 seeds.

Findings (done).

- On GR00T, conditioning lifts seen success on every training seed: 9–12 of 20 per seed for `kv_adaln`, against 2–4 for the control.
- `kv_adaln`'s held-out gain also holds on every seed (1–6 of 20, against 0). `kv`'s does not: one seed scores 0.
- AdaLN's held-out effect is replicated three times: ACT on atomic clips (0→6), ACT on full runs (0→7), and GR00T (0→11).
- Conditioning without stage 1 hurts (ACT `scratch_kv`: 21 → 2, $p < 0.001$).

Why it works (done).

- *It is information, not capacity.* The identical network trained on another scene's entities scores 8 (vs. 35, $p < 0.001$).
- *Query-dependent composition carries it.* A pooled scene vector alone does not beat the control (12 and 8 vs. 9). Adding the KV path takes it from 12 to 31 ($p = 0.001$).
- *Pose noise and appearance.* Performance holds up to ArUco-level pose noise and drops to the control's level only when a quarter of the entities are missing. Under unseen textures and lighting the RGB control collapses (9→2, $p = 0.039$), while the conditioned arms keep 24–25 of 60.
- *Precision.* The miss is horizontal position (˜9 mm for every arm). AdaLN is more precise in height (1.5 mm vs. 3.5 mm). This motivates relative geometry (Sec. III-C).

Open: why GR00T but not $\pi_{0.5}$/SmolVLA? GR00T's branch is ˜150M parameters against ˜1M for the matched LoRA branch, so the ratio is the same but the absolute size is not. Decide whether to test a larger branch with a capacity-matched control, or to report the backbone dependence as a finding.

VI. RQ2: DOES RELATIONAL COMPOSITION HELP?

Claim. (Hypothesis.) On top of per-entity retrieval, pairwise relative geometry improves multi-object success and grasp precision. Colour is necessary when the goal is defined by colour.

Have. Motivation only: the controls above, plus the horizontal grasp error that no absolute-pose arm reduces.

Need. Implement `kv_rel` and its controls in `policies/conditioning/`. Run GR00T on restock first (3 arms × 3 seeds), then RoboTwin; then the `+c` arms on the two colour tasks.

Pre-registered readings.

- A relational gain counts only if `kv_rel` beats both `kv_mlp` and `kv_rel_shuf`, on every seed.
- The colour tasks: `kv` without colour should not exceed the control, which also checks that the entity branch is not inert.

VII. RQ3: GENERALISATION ACROSS OBJECT COUNT

Claim. GR00T conditioning carries over to two objects strongly and to four objects marginally. Per-entity `kv` completes four-object scenes; pooled AdaLN reaches fewer first transfers than the control. ACT completes no four-object scene at all.

Have. GR00T and ACT, existing arms, 2/3/4 objects (Table VI).

Need. `kv_rel` and its controls at 2 and 4 objects. Optional: RoboTwin count pairs (stack_blocks_two/three, stack_bowls_two/three).

- GR00T, against `rgb_cont`:
  - two objects: 14→47 for `kv` and 14→37 for `kv_adaln` (both $p < 0.001$);
  - four objects: `kv` 0→9 ($p = 0.004$), on every seed (1–5 of 20);
  - at four objects `kv_adaln` reaches a first transfer less often than the control, 37→19 ($p = 0.002$), and `kv` beats it there (31 vs. 19, $p = 0.043$).
- Reading, untested: the pooled summary is fitted on three-object scenes, while per-entity retrieval is not tied to a

TABLE II
RESTOCK, THREE OBJECTS, PAIRED AGAINST THE BUDGET CONTROL (60 MATCHED EPISODES PER CELL; EXACT MCNEMAR p IN PARENTHESES).

Backbone	Arm	Success, seen	Success, held-out	≥1 transfer, seen
ACT	`rgb_cont`	21	0	39
	`kv`	22 (1.0)	0	37 (0.85)
	`kv_adaln`	22 (1.0)	**7 (0.016)**	33 (0.35)
	`kv_tokens`	26 (0.44)	0	41 (0.85)
GR00T N1.7	`rgb_cont`	9	0	36
	`kv`	**31 (<0.001)**	**7 (0.016)**	40 (0.57)
	`kv_adaln`	**35 (<0.001)**	**11 (0.001)**	**49 (0.019)**
$\pi_{0.5}$	`rgb_cont`	0	0	9
	`kv`	1 (1.0)	0	21 (0.023)
	`kv_adaln`	2 (0.50)	0	8 (1.0)
SmolVLA	`rgb_cont`	0	0	13
	`kv`	4 (0.13)	0	17 (0.54)
	`kv_adaln`	3 (0.25)	1 (1.0)	12 (1.0)
RoboTwin (5)	all arms	?	–	?

TABLE III
GR00T: INFORMATION, COMPOSITION AND ROBUSTNESS (SUCCESS OF 60).

Arm	Seen	Held-out	Seen, visual shift
`rgb_cont`	9	0	2
`kv_adaln`, shuffled entities	8	3	–
scene vector only, mean	8	1	–
scene vector only, attention	12	3	–
`kv`	31	7	25
`kv_adaln`	35	11	24
Entity noise at evaluation, seen success (`kv` / `kv_adaln`)			
none	31 / 35		
2 mm, 2°	32 / 28		
5 mm, 2°	28 / 33		
10 mm, 5°	24 / 23		
5 mm, 2°, 10% dropout	25 / 24		
5 mm, 2°, 25% dropout	12 / 10		

TABLE IV
RQ2 ARMS. EACH ISOLATES ONE ADDITION OVER `kv`.

Arm	Change vs. `kv`	Controls for
`kv_mlp`	per-entity MLP, no pairs, same params	capacity
`kv_rel`	pairwise u_{ij}, Eq. (3)	–
`kv_rel_shuf`	u_{ij} from another scene	information
`kv+c`, `kv_rel+c`	+ colour columns	semantics

count. *Prediction:* `kv_rel`, with mean aggregation, is at least `kv` at four objects.

- ACT: 0 of 1380 four-object episodes succeed, across every arm and corpus, even at the full 800 steps. ACT is reported as a floor and does not carry RQ3.

Fig.: success vs. object count (2/3/4), one line per arm, seed range as error bars.

VIII. LIMITATIONS

- Ground-truth entities. The noise study is a partial answer; there is no perception pipeline.
- Simulation only.

TABLE V
RQ2 RESULTS: RESTOCK SUCCESS OF 60, MEAN ROBOTWIN SUCCESS, MEDIAN HORIZONTAL GRASP ERROR.

Arm	Seen	Held-out	RoboTwin	Err. [mm]
`rgb_cont`	9	0	?	?
`kv`	31	7	?	?
`kv_mlp`	?	?	?	?
`kv_rel`	?	?	?	?
`kv_rel_shuf`	?	?	?	?

TABLE VI
TRAINED ON 3 OBJECTS. SUCCESS OF 60 AT 2 / 3 / 4 OBJECTS; IN BRACKETS, EPISODES WITH AT LEAST ONE TRANSFER AT 4 OBJECTS.

	Arm	2	3	4	[4: ≥1]
GR00T	`rgb_cont`	14	9	0	[37]
	`kv`	**47**	31	**9**	[31]
	`kv_adaln`	**37**	35	4	[**19**]
	`kv_rel`	?	?	?	[?]
ACT	`rgb_cont`	18	21	0	[22]
	`kv`	18	22	0	[14]
	`kv_adaln`	16	22	0	[18]

RA-L reviewers will likely ask for a real-robot result. The 2–5 mm robustness makes a fiducial-tracked real-robot cell plausible; decide whether to scope one in.

- The restock objects differ only in size.
- GR00T, $\pi_{0.5}$ and SmolVLA rollouts are unseeded draws, so individual scenes do not reproduce. The paired tests remain valid.
- Seed variance is large. Same-recipe retrains differ at $p = 0.008$.

IX. CONCLUSION

After RQ2/RQ3.

EXECUTION PLAN (REMOVE BEFORE SUBMISSION)

1) **RQ2 on restock, GR00T** (cheapest; run first). `kv_mlp`, `kv_rel` and `kv_rel_shuf` × 3 seeds (9 stage-2

jobs). Roll out on seen, held-out, 2-object and 4-object scenes (36 rollouts). Then the same arms on ACT.

2) **RoboTwin infrastructure**, per task:
   - an `ObjectBinding` (only `place_container_plate` has one today);
   - an oracle probe on a few seeds before scaling;
   - a `plan_success` check;
   - the export;
   - a closed-loop evaluator for built-in tasks, which does not exist yet;
   - the oracle ceiling.
3) **RoboTwin matrix.** 5 tasks $\times$ {ACT, GR00T} $\times$ {`rgb`, `rgb_cont`, `kv`, `kv_adaln`, `kv_rel`} $\times$ 3 seeds, which is about 150 training jobs. Add the `+c` arms on the two colour tasks, about 24 more.
4) **Colour entity format.** A new export and re-run, kept separate from the other changes so it cannot confound them.

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